AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 16 · STUDENT LESSON

# Ablation studies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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## Before you start

Recall Multiple comparisons. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Explain the mechanism and terminology of ablation studies. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of **ablation** studies.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **ablation** studies

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Multiple comparisons. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

### This week's end goal

Explain **ablation** studies using the records and boundaries of this fictional investigation.

## Notice the evidence

> Investigate the supplied Ablation studies case. Separate observations, given facts and assumptions before choosing an explanation.

**Ablation** removes a component to investigate its contribution under a controlled comparison. **Interaction**s can make contributions depend on other components.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

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STUDENT LESSON · 3 OF 18

## See how it works

> Follow the mechanism for Ablation studies. Connect each labelled input, deciding step and result.

Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals.

### Ablation studies — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Full system8/10, without retrieval5/10, without validation7/10; the last permits an unauthorised action. | Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals. | Drops3 and1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

> Compare the supplied cases for Ablation studies. Identify the detail that changes the conclusion.

### Compare a familiar case with a changed case

Full system8/10, without retrieval5/10, without validation7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

### Change one thing in the pictured case

#### Change one thing

Change: Remove the filter and also change the model version.

#### Predicted result

**Prediction**: The comparison no longer isolates the filter.

#### Why the result follows

Reason: Two simultaneous changes provide alternative explanations.

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STUDENT LESSON · 5 OF 18

## Words that unlock the idea

Use the vocabulary of Ablation studies to explain a concrete decision, rather than only memorising definitions.

| Word | Meaning |
| --- | --- |
| Ablation | Removing or disabling a component in a comparison |
| Conditional contribution | An effect given the other components |
| Interaction | An effect depending on another factor's setting |

**Use two terms to explain a step in this week's example.**

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STUDENT LESSON · 6 OF 18

# Follow a worked example

> Follow the worked trace for Ablation studies, predict a step, then compare with its explanation.

Full system8/10, without retrieval5/10, without validation7/10; the last permits an unauthorised action.

Drops3 and1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

### Worked explanation · Ablation studies

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Ablation studies

Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.

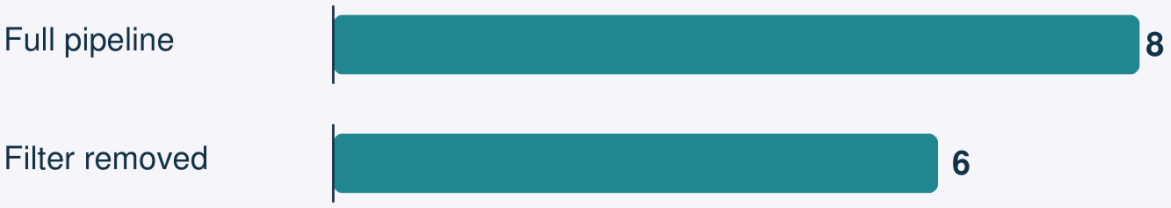
### The facts we will use

> **Supplied facts**
>
> Full pipeline: 8/10 correct. Filter removed: 6/10 on the same cases, with all other settings fixed.

AI ACADEMY / AI-7/16

## Relevance-filter ablation

correct answers out of 10 shared cases




Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/16

# Ablation studies

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Name the component

The relevance filter is the single declared change.



## 2. Run the ablation

Remove only the filter and preserve cases and scoring.



## 3. Compare outcomes

Correctness falls from 80% to 60%, two cases in this set.

**CHANGE ONE THING**

Remove the filter and also change the model version.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Name the component | The relevance filter is the single declared change. |
| 2. Run the ablation | Remove only the filter and preserve cases and scoring. |
| 3. Compare outcomes | Correctness falls from 80% to 60%, two cases in this set. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

### What this example does not establish

#### Keep the stated limit

**Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

### Keep the stated limit

Read the labelled picture: Calculate the filter **ablation**'s two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

### Keep the stated limit

Change one thing: Remove the filter and also change the model version. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Ablation studies claim and explain a repair using the supplied evidence.

Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

### Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Locate the earliest unsupported step

A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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STUDENT LESSON · 8 OF 18

## Practise together

> Practise Ablation studies with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Ablation** studies". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for **Ablation** studies. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

> Calculate the filter **ablation**'s two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

## Run the investigation

> Investigate Ablation studies. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

### Predict, investigate and preserve the evidence

Full system8/10, without retrieval5/10, without validation7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or "not run" | |

| Record | Your evidence |
| --- | --- |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Ablation studies. Explain why each step is justified.

**Explain → apply → challenge**

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

## Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **ablation** studies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **ablation** studies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

### Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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STUDENT LESSON · 12 OF 18

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

# Study a complete case

> Revisit the complete Ablation studies case for supported practice. Repeated examples do not become fresh assessments.

## Complete evidence record

### Ablation studies — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Full system8/10, without retrieval5/10, without validation7/10; the last permits an unauthorised action. | Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals. | Drops3 and1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

**My complete trace of the worked case**

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STUDENT LESSON · 14 OF 18

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Apply a new case independently

Full12/15, without A9/15, without B11/15. Compute sample drops and state **interaction** limit.

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

Full12/15, without A9/15, without B11/15. Compute sample drops and state **interaction** limit.

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

> Use this map to navigate Ablation studies, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter’s contribution.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of ablation studies | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Ablation studies without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **ablation** studies

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? Correctness falls from **80%** to **60%**, two cases in this set.

1. Name the component

2. Run the **ablation**

3. Compare outcomes

**My selected conclusion, deciding fact and explanation**

**Hint · try first**

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare outcomes Correctness falls from **80%** to **60%**, two cases in this set. An **ablation** probes a component by removing or changing it while holding the rest fixed.

It supports a scoped comparison, not a claim that the component always helps.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Remove the filter and also change the model version. Which conclusion is supported by the supplied facts?

1. The comparison no longer isolates the filter.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The comparison no longer isolates the filter. Two simultaneous changes provide alternative explanations.

#### Repair a missing building block

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

#### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

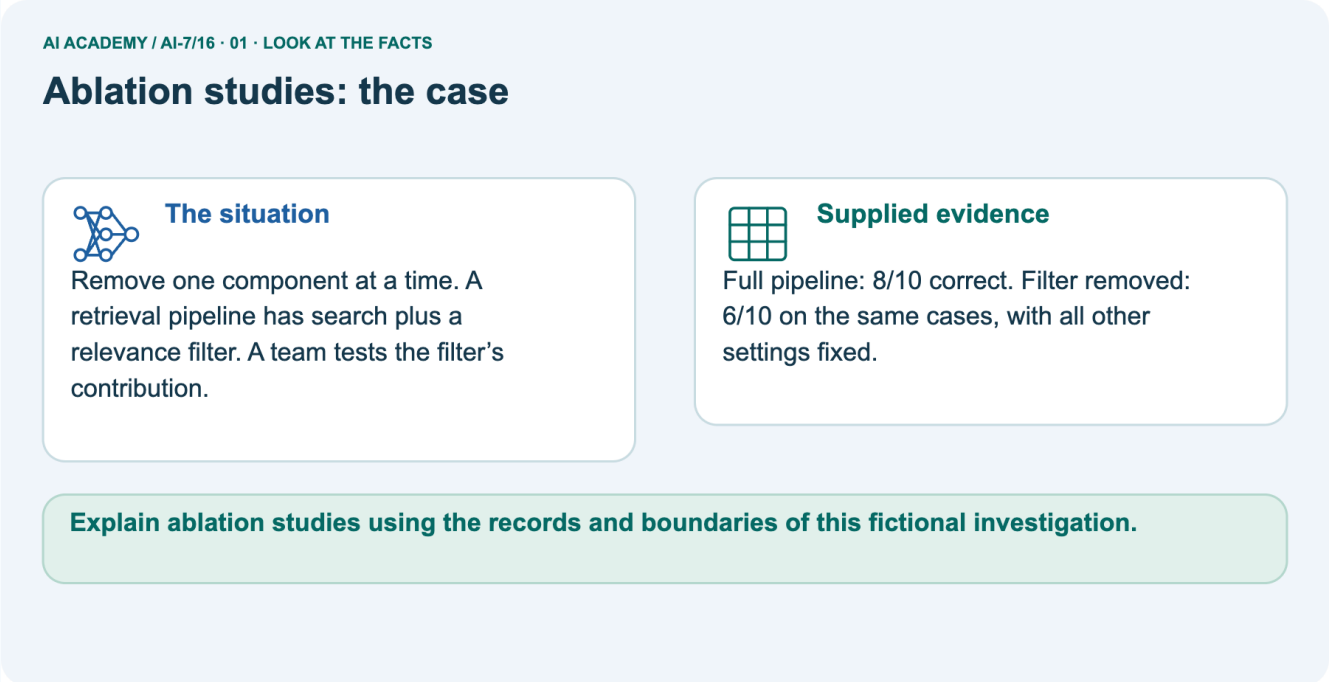
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

# Visual concept journey

> Explain each Ablation studies graphic using its labels, deciding evidence and relevant limitation.

### 01. Ablation studies: the case



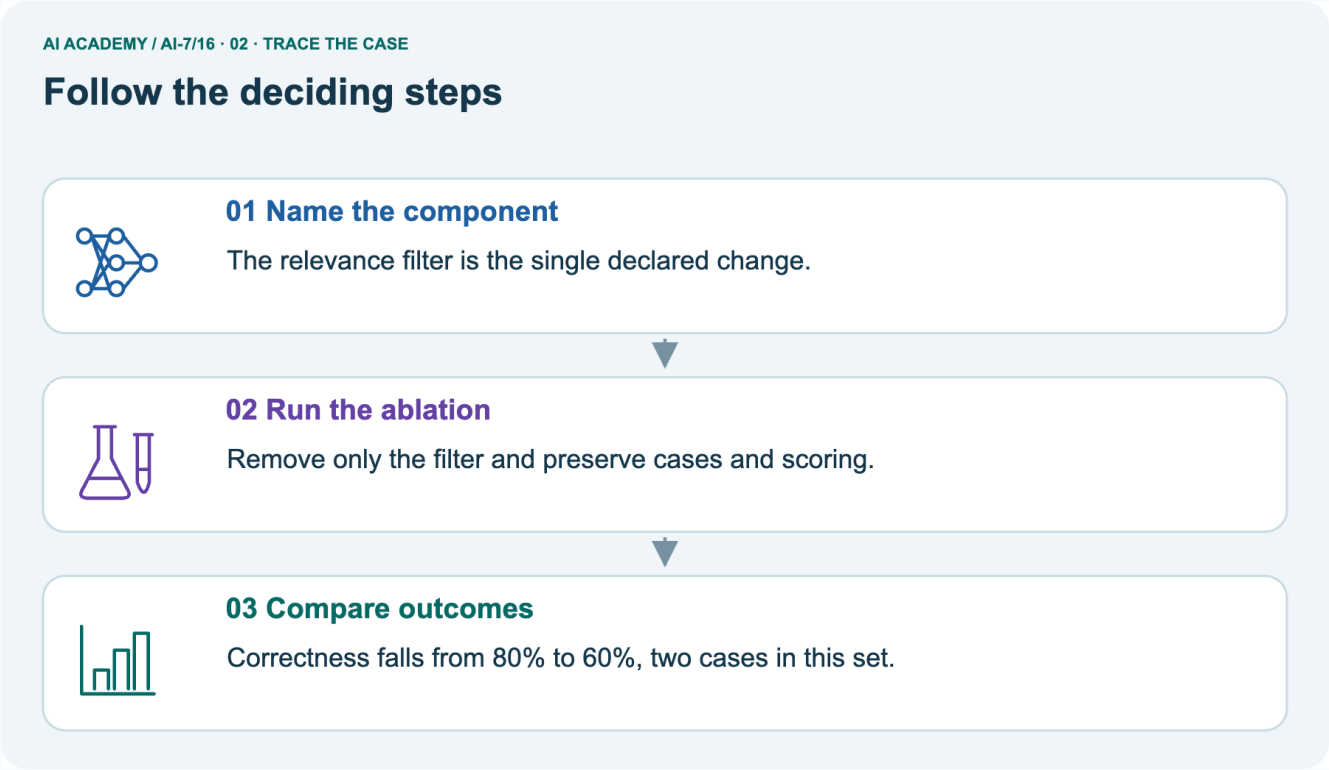

Follow the labelled picture with the matching explanation. Open the vector diagram

Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution. Supplied facts: Full pipeline: 8/10 correct.

Filter removed: 6/10 on the same cases, with all other settings fixed.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

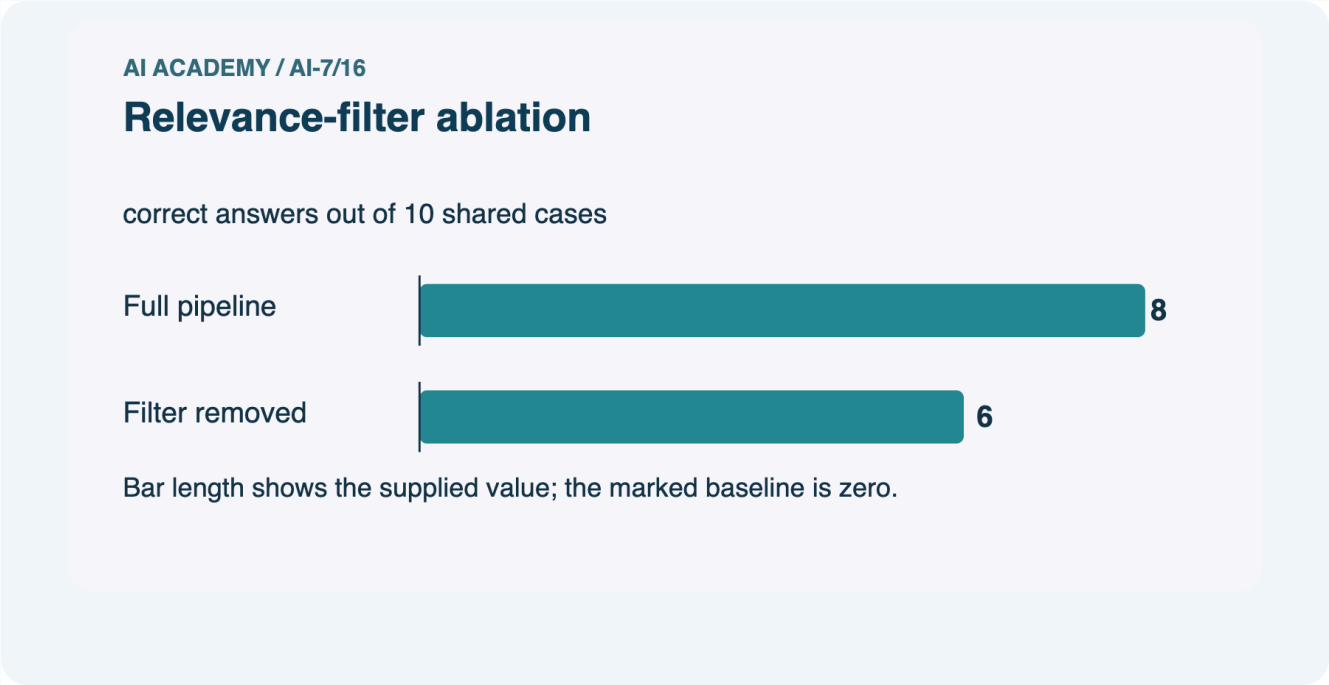



Follow the labelled picture with the matching explanation. Open the vector diagram

Name the component: The relevance filter is the single declared change. → Run the **ablation**: Remove only the filter and preserve cases and scoring. → Compare outcomes: Correctness falls from **80%** to **60%**, two cases in this set.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. Relevance-filter ablation

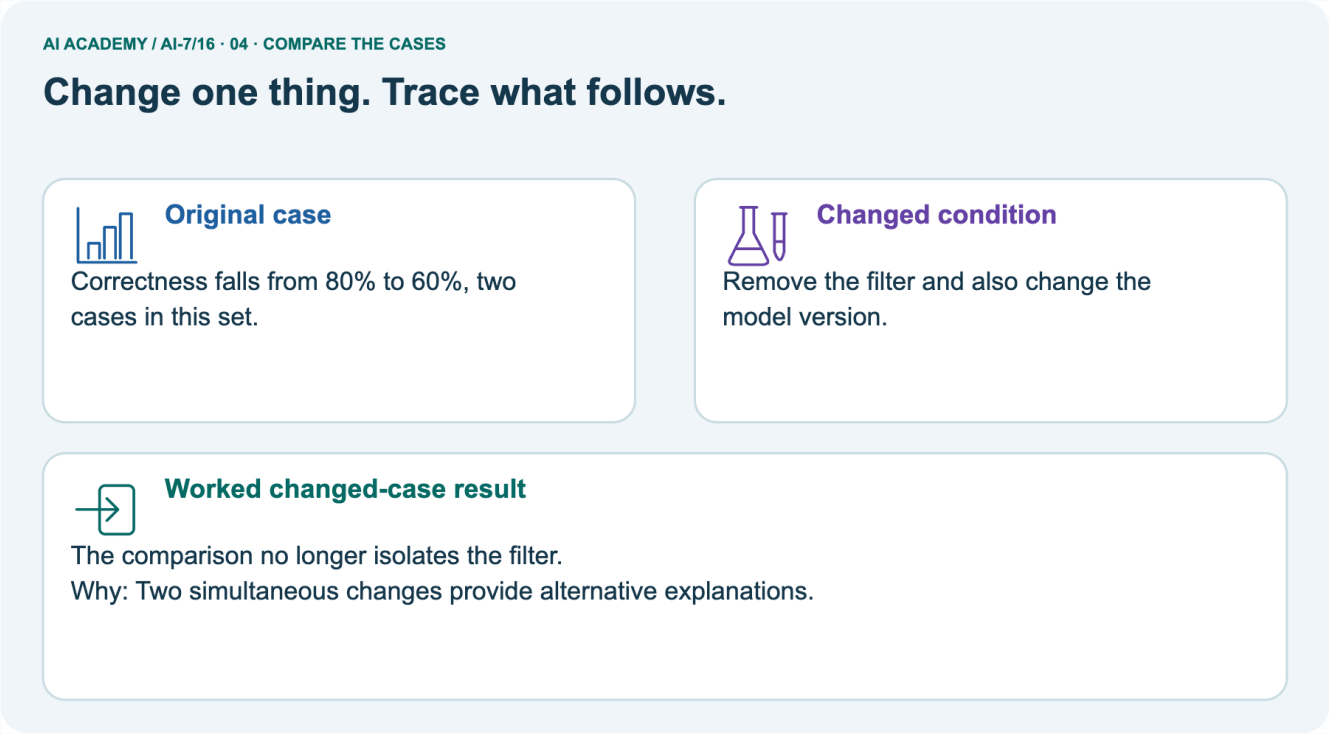



Follow the labelled picture with the matching explanation. Open the vector diagram

Relevance-filter **ablation**. Units: correct answers out of 10 shared cases. Full pipeline: 8. Filter removed: 6.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



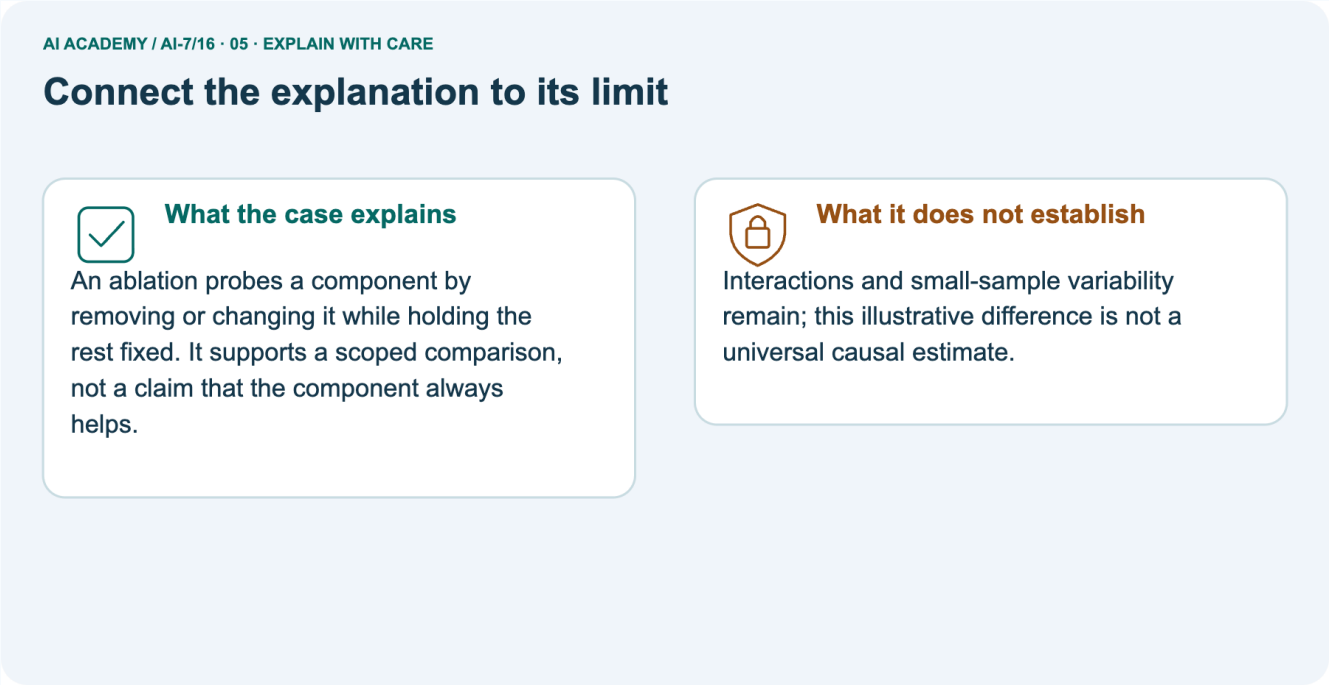

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correctness falls from **80%** to **60%**, two cases in this set. Changed condition: Remove the filter and also change the model version. Expected result: The comparison no longer isolates the filter.

Reason: Two simultaneous changes provide alternative explanations.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



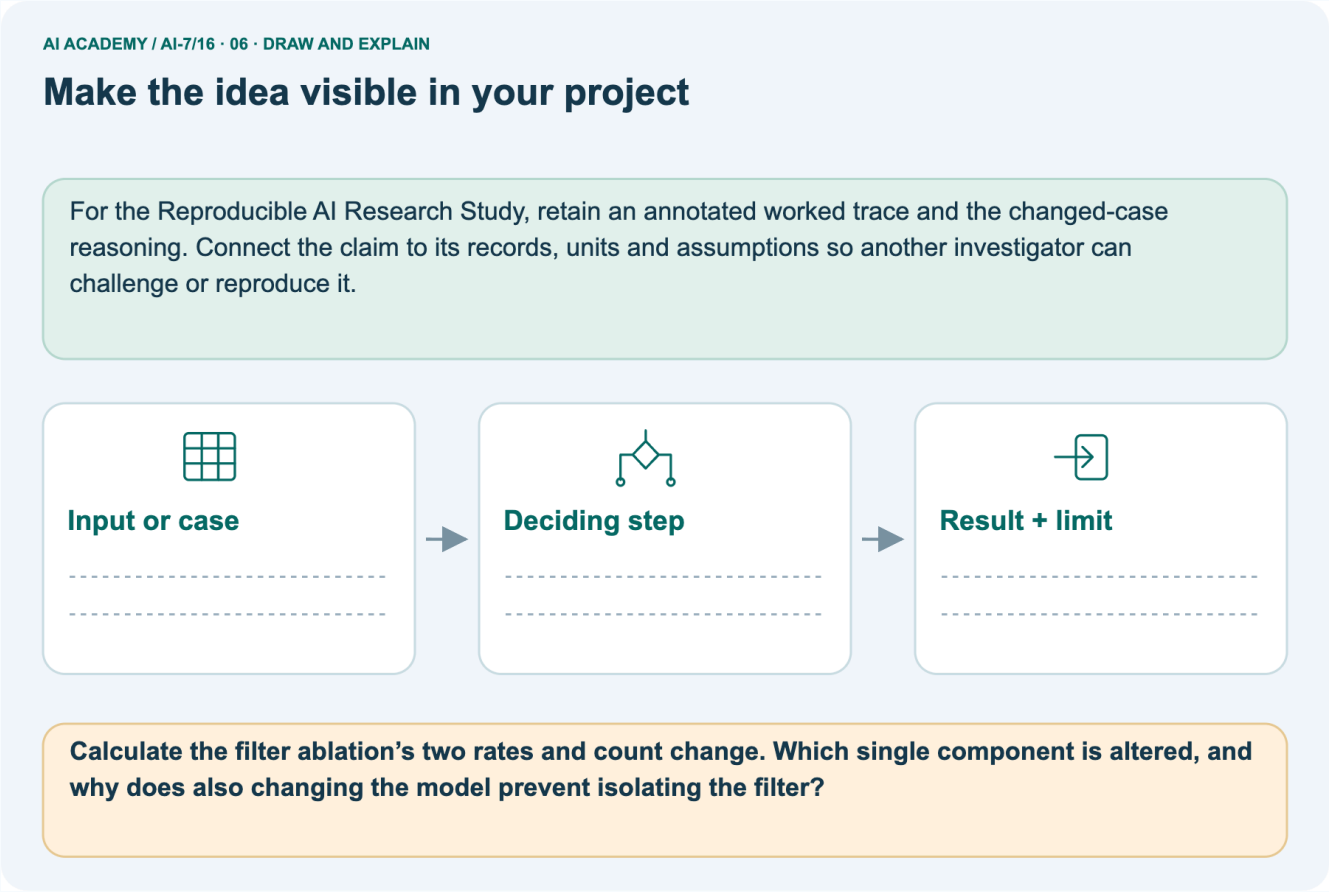

Follow the labelled picture with the matching explanation. Open the vector diagram

An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

Limit: **Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the filter **ablation**'s two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.